

Benchmark 4-quadrant - wider replication

Date: 2026-05-26

Models: google/gemma-3-4b-it, meta-llama/llama-3.2-3b-instruct, qwen/qwen-2.5-7b-instruct

Observations: 25 (gemma 10 + llama 10 + qwen 5 partial)

Headline

Quadrant	Pass rate	Avg iters	Tokens/pass	ms/pass
Q1 raw 1-shot	0/25 (0%)	1.0	22,387	817,437
Q2 simplicio 1-shot	16/25 (64%)	1.0	1,093	14,797
Q3 raw + loop	11/25 (44%)	4.0	7,154	106,382
Q4 simplicio + loop	19/25 (76%)	2.44	1,914	24,170

Decomposition (delta in points)

Q2 - Q1 prompt effect (1-shot)	+64 pts
Q3 - Q1 loop effect (raw)	+44 pts
Q4 - Q3 prompt effect inside loop	+32 pts
Q4 - Q2 loop effect with simplicio	+12 pts
Q4 - max(Q2,Q3) composition gain	+12 pts
Q4 - linear synergy vs stacking	-32 pts

Hypothesis verdicts (threshold |delta|>=5 pts)

Loop alone closes the gap (simplicio unnecessary once you loop)	+32	REJECTED
Simplicio alone is enough (loop is overkill)	+12	REJECTED
Gains stack linearly (no synergy)	-32	REJECTED

Per-model breakdown

Model	Q1	Q2	Q3	Q4
google/gemma-3-4b-it	0/10 (0%)	7/10 (70%)	4/10 (40%)	8/10 (80%)
meta-llama/llama-3.2-3b-instruct	0/10 (0%)	5/10 (50%)	4/10 (40%)	6/10 (60%)
qwen/qwen-2.5-7b-instruct	0/5 (0%)	4/5 (80%)	3/5 (60%)	5/5 (100%)

Note: partial run - wide bench killed mid-execution; reconstructed from log.